MILITARY SPECIFICATION

PANELS, FIBER - REINFORCED PLASTIC FOR RAPID RUNWAY REPAIR

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1. Scope. This specification covers bolt-together fiberglass-reinforced plastic (FRP) panels made by the resin transfer molding (RTM) method, and fasteners for use as a cover for backfilled bomb craters within airfield pavements or as an expedient runway surfacing.

1.2 Classification. The FRP panels shall be of the following types as specified (see 6.2):

Type I	- Full-size	- 216.0 x 80.0 inches.
Type II	- Left Half-size	- 112.0 x 80.0 inches.
Type III	- Right Half-size	- 104.0 x 80.0 inches.
Type IV	- Ramp panel	- 208.0 x 36.0 inches.
Type V	- Center panel	- 216.0 x 40.0 inches.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5680

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-C-3774 - Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-C-81346 - Compound, Deck Covering, Nonslip, Lightweight, for Aluminum Alloy Landing Mats.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D638 - Tensile Properties of Plastics, Test Method for
ASTM D790 - Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials, Test Method for

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Five types of FRP panels, full-size, left half-size, right half-size, ramp, and center, are described in this specification. Also, the bolts and bushings required to assemble the panels to form a cover are described.

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3.2 First article. When specified (see 6.2), the contractor shall furnish one complete panel of each type ordered for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.3.1 FRP material. FRP panels shall be fabricated from fiberglass, polyester or vinyl ester resin, and high strength aramid fabric, such as style 1350 Kevlar-49 fabric or equivalent (tensile strength 500,000 pound-force per square inch (psi) minimum).

3.3.2 Bushing material. Unless otherwise specified (see 6.2), bushings shall be made from resulfurized carbon steel (AISI C1144) meeting the mechanical properties given in Table I.

TABLE I
MECHANICAL PROPERTIES OF CARBON STEEL (AISI C1144)

Tensile Strength (psi) (minimum)	132,000
Minimum Yield Point (psi) (minimum)	100,000
Percent Elongation (in 2 inches) (minimum)	10
Percent Reduction of Area (minimum)	35
Average Brinnell Hardness	269

3.4 Construction of FRP panels. Unless otherwise specified (see 6.2), the FRP panels shall be fabricated as shown in figures 1 through 9 using the RTM method. The panels shall contain three plies of 40 ounce per square yard woven roving fiberglass, and three plies of 2 ounce per square foot continous strand fiberglass. The cured resin, when used in conjunction with the fiberglass at 55:45 (approximate) resin-to-glass ratio by weight shall produce a laminate having the mechanical properties as shown in Table II when tested as specified in section 4.6. Panel edges shall be reinforced with two 14-inch wide plies of high strength aramid fabric (tensile stregth, impregnated, 500,000 psi minimum) as shown in figures 1 and 2. It is imperative that the upper layer of woven aramid on the upper joint continue underneath the flange of the upper bushing, see figures 8, 9, and 15. Also, an extra layer of continuous strand fiberglass shall go beneath the lower bushing to keep it inplace, see figure 15.

TABLE II
MECHANICAL PROPERTIES OF FRP PANELS

PROPERTY	MEAN STRENGTH (psi)	STANDARD DEVIATION (psi)
Tensile Strength (Min)	15,000	1,800
Elastic Modulus, Tension (Min)	$1x10^6$	--
Flexural Strength (Min)	29,000	3,000

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3.4.1 Full-size panels. Panels shall be 3/8" (nominal) thick. Panel edges shall be 8 inches wide and recessed to 1/4-inch thickness on top edge and 1/8-inch thickness on the bottom edge. Two adjecent edges of a panel shall have the recess at the panel bottom, the other two edges shall be recessed at the panel top. A plan view of the basic panel is shown in figure 3. Edge details are shown in figures 1 and 2. Bushing locations along panel edges are shown in figure 3.

3.4.2 Half-size panels. Half-size panels shall be made one half of a full-size panels. Equally divide the bushing hole pattern of the top recessed edge, see figures 4 and 5.

3.4.3 Center panels. Center panels shall be as shown in figure 6. Bushing locations are shown in figure 6.

3.4.4 Ramp panels. Ramp panels shall be as shown in figures 7, 8 and 9. Bushing locations are shown in figure 7.

3.5 Weight. Maximum weight of the panels shall be as given in Table III.

Table III
MAXIMUM WEIGHT OF PANELS

Description	Maximum weight, lbs
Full-size panel	300
Half-size panel	150
Center panel	150
Ramp panel	100

3.6 Upper bushing. The upper bushing, shown in figure 10, shall be molded in the locations shown in the applicable figures.

3.7 Lower bushing. The lower bushing, shown in figure 11, shall be molded in the locations shown in the applicable figures.

3.8 Anchor bushing. The anchor bushing, of the type specified (see 6.2), shown in figures 12 and 13, shall be molded in the locations shown in figure 7.

3.9 Bolts. Unless otherwise specified (see 6.2), the contractor shall supply sufficient number of corrosion resistent hex head steel bolts, as shown in figure 14, for jointing the panels. The bolts shall be threaded over their entire length, have a minimum tensile strength of 120,000 psi and have treads compatible with the lower bushings.

3.10 Dimensions. Dimensions shall be as shown in the appropriate figure.

3.11 Tolerances. The contractor shall control dimensional tolerances to ensure that the completed FRP panels can be assembled in a random order and jointed with bushings and hex head bolts.

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3.12 Nonskid coating. Unless otherwise specified,(see 6.2), each panel shall be coated on the top wearing surface with a nonskid coating in accordance with MIL-C-81346, except for the 8-inch wide bottom edges, The thickness of the nonskid surface shall be not less than 5 and not more than 10 mils thick.

3.13 Interchangeability. All panels of the same classification furnished options shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.14 Workmanship. Workmanship of the FRP panels, and bushings shall be of the quality prevailing among manufactures normally producing similar items to those specified, and the panels shall be free from the following defects:

a. Delamination.
b. Dry spots.
c. Voids.
d. Blisters.
e. Crazing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on each type of panel ordered. This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract. The panels produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of

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Goverment representatives. The purpose of the inspection shall be to determine panel conformity with the contract. Acceptance of the first production panel shall not consitute a waiver by the Government of its rights under the provisions of the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete panel of the type specified. All panels of the same type offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4 Examination. Each FRP panel and bushing shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests. Twelve FRP specimens shall be prepared from each batch of material for testing. Four specimens shall be used for each of these tests. Failure to comply with the mechanical properties as listed in Table II shall be cause for rejection of the panels. All tests shall be performed on specimens without the aramid fabric.

4.5.1 Tensile strength. In accordance with ASTM D638.

4.5.2 Elastic modulus, tension. In accordance with ASTM D638.

4.5.3 Flexural strength. In accordance with ASTM D790.

4.5.4 Dimensional accuracy. An overall check of the dimensional accuracy shall be made by joining the panels in a random order to form a cover, as shown in figure 16, of the size specified (see 6.2). The joining shall be complete with bushings and bolts. Binding, misalignment or the inability to join the panels in this manner to form a cover shall be cause for rejection.

4.5.5 Interchangeability test. When panels pass the test of 4.5.4, disassemble cover and reassemble panels in different sequence.

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4.6 Packaging inspection. The preservation, packing, and marking of the panels shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Packing. Packing shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Level A panels. The panels shall be packed in crates conforming to MIL-C-3774, type as specified below. The weight of each crate shall be at the contractor's option: however, the weight limits as specified in MIL-C-3774 shall not be exceeded. Only panels of similar classification (i.e., panels of the same type) shall be packed together in a crate. Panels shall be laid flat, not on edge, in the crate. When specified (see 6.2), a full shroud shall be provided in accordance with the appendix to MIL-C-3774.

Type of panel	Type of crate
I	II
II, III	I OR II
IV	II
V	II

5.1.1.2 Level A bolts. Individual preservation or packaging of bolts is not required. Bolts shall be packed in boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2.

5.1.2 Level C.

5.1.2.1 Level B panels. The panels shall be packed in a unitized load configuration. Only panels of similar classification (i.e., panels of the same type) shall be unitized. Panels shall be laid flat, not on edge. Panels shall be packed in a manner to prevent deterioration and damage and insure arrival at destination in a satisfactory condition.

5.1.2.2 Level B bolts. Individual preservation or packaging of bolts is not required. Bolts shall be packed in boxes conforming to PPP-B-636, class domestic; PPP-B-640, class 1; PPP-B-585, class 1; PPP-B-601, domestic type, or PPP-B-621, class 1.

5.2 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers bolt-together FRP panels made by the RTM method, and fasteners for use as a cover for backfilled bomb craters within airfield pavements or as an expedient runway surfacing.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and number of panels required (see 1.2).
c. If reference documents are other than those specified (see 2.1.1).
d. When first article is required for inspection and approval (see 3.2).
e. If bushings are to be made from different material than what is specified (see 3.3.2).
f. If panels are to be fabricated different than what is specified (see 3.4)
g. Type of anchor bushing required (see 3.8).
h. If the contractor shall supply the hex bolts (see 3.9)
i. If the nonskid coating is to be other than specified (see 3.12)
j. Size of cover to be assembled to check the overall dimensional accuracy of the panels (see 4.5.4).
k. Level of packing required (see 5.1).
l. When a full shroud is required (see 5.1.1.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete panel of each type ordered. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

Preparing Activity

Navy - YD

(Project 5680-N136)

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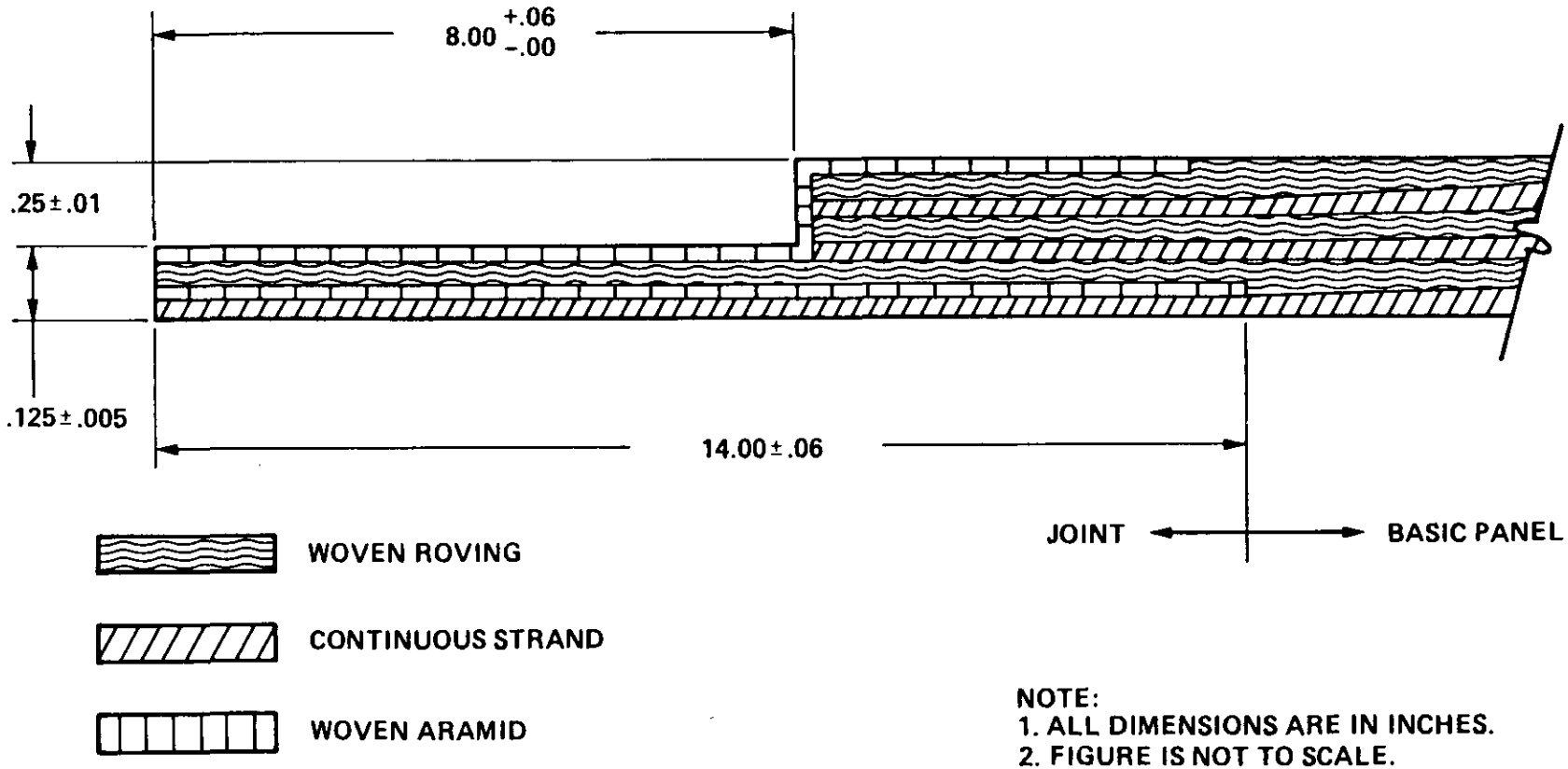
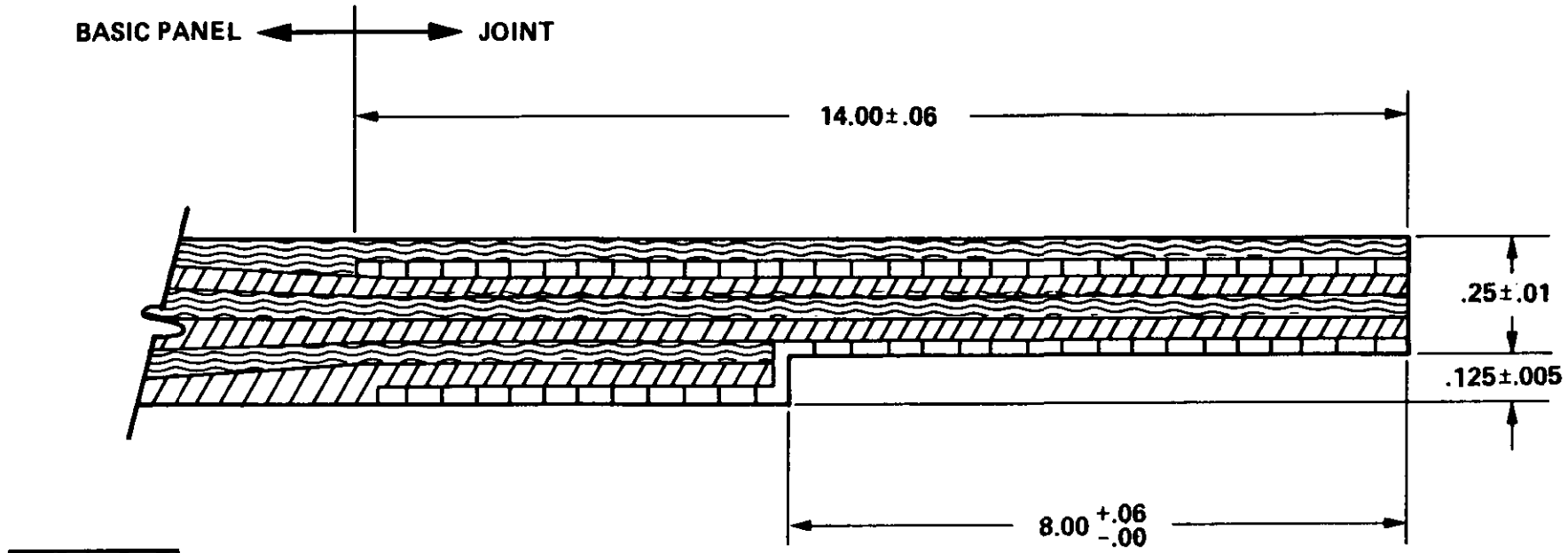


FIGURE 1. Lower joint, Section A-A.

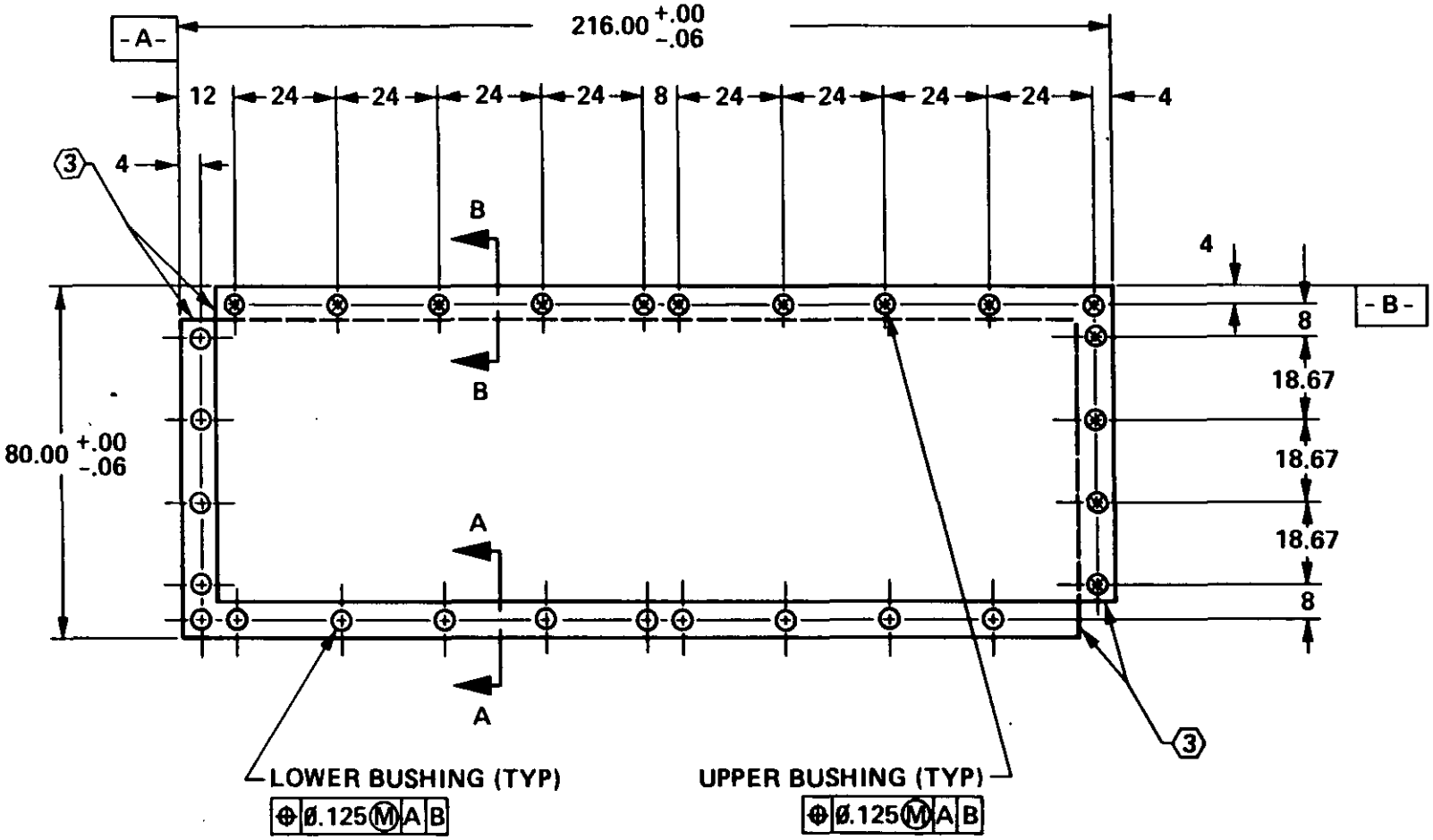
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NOTE:
1. ALL DIMENSIONS ARE IN INCHES.
2. FIGURE IS NOT TO SCALE.

FIGURE 2. Upper joint, Section B-B.

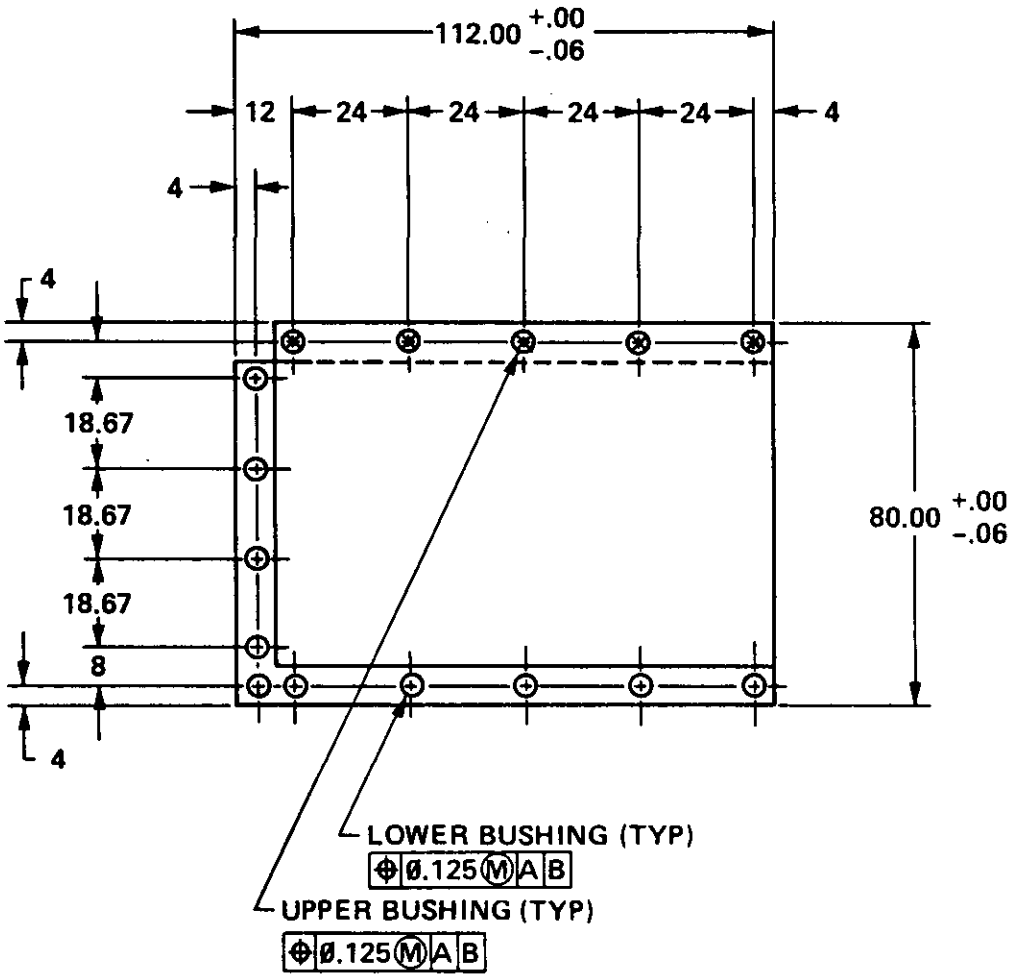
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NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. UNTOLERANCED DIMENSIONS ARE BASIC.
3. REDUCE AN ADDITIONAL 0.1" FROM THESE EDGES FOR ASSEMBLY TOLERANCE.
4. FIGURE IS NOT TO SCALE.

FIGURE 3. FRP airfield matting panel (RTM method), Plan View.

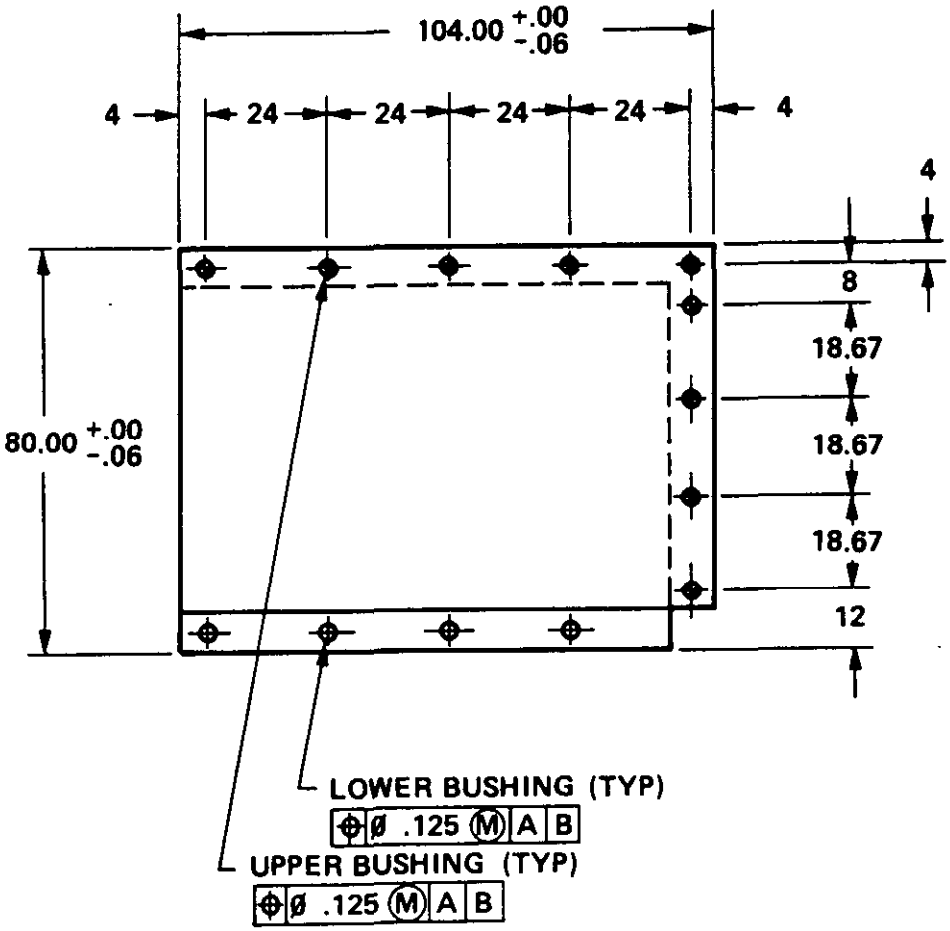
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NOTES:
1. LEFT HALF PANEL FITS ON THE RIGHT SIDE OF THE COVER AND IS 8" LONGER THAN THE RIGHT HALF PANEL.
2. ALL DIMENSIONS ARE IN INCHES.
3. FIGURE IS NOT TO SCALE.

FIGURE 4. Left half panel, Plan View.

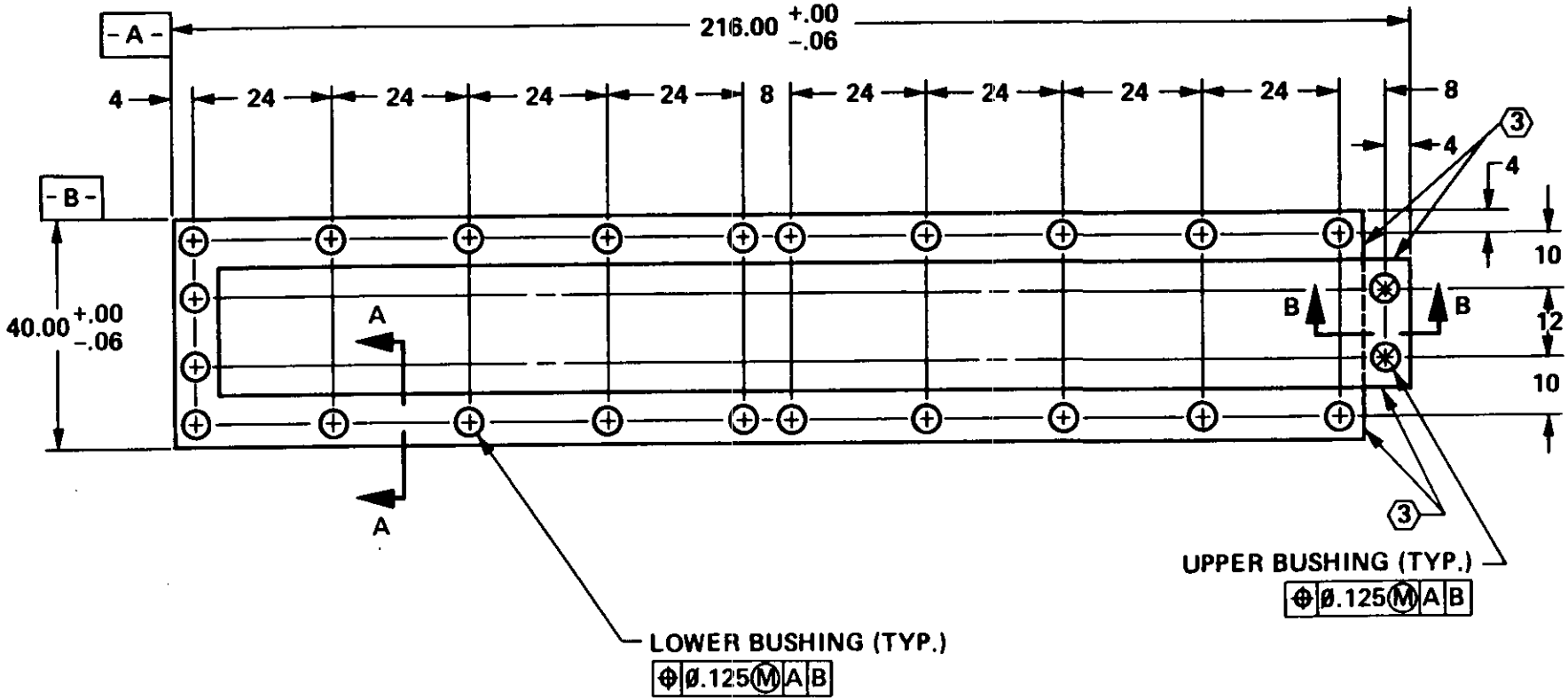
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NOTES:
1. RIGHT HALF PANEL FITS ON THE LEFT HAND SIDE OF THE COVER AND IS 8" SHORTER THAN THE LEFT HALF PANEL.
2. ALL DIMENSIONS ARE IN INCHES.
3. FIGURE IS NOT TO SCALE.

FIGURE 5. Right half panel, Plan View.

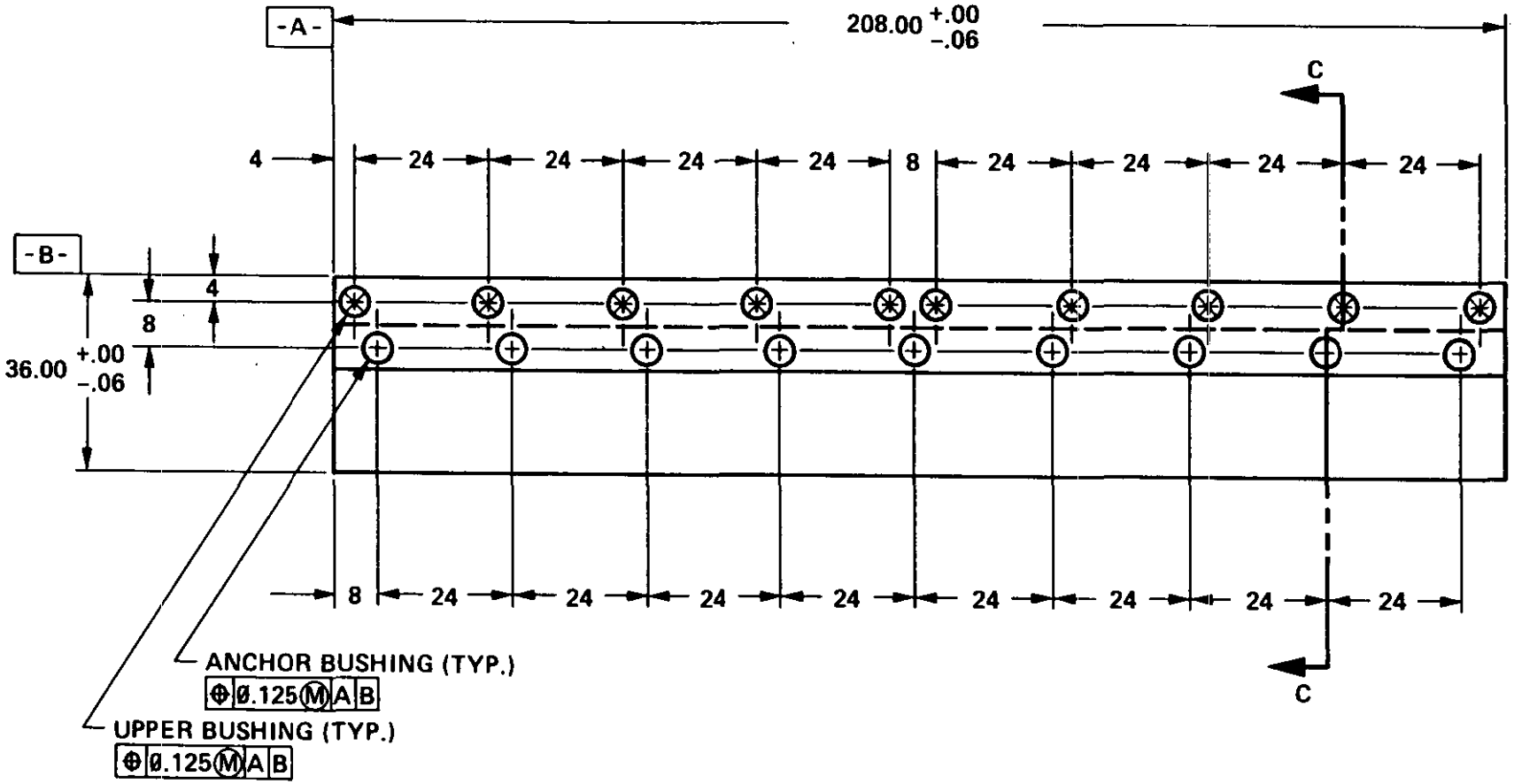
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NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. UNTOLERANCED DIMENSIONS ARE BASIC.
3. REDUCE AN ADDITIONAL 0.1" FROM THESE EDGES FOR ASSEMBLY TOLERANCE.
4. FIGURE IS NOT TO SCALE.

FIGURE 6. FRP airfield matting panel (RTM method), center panel.

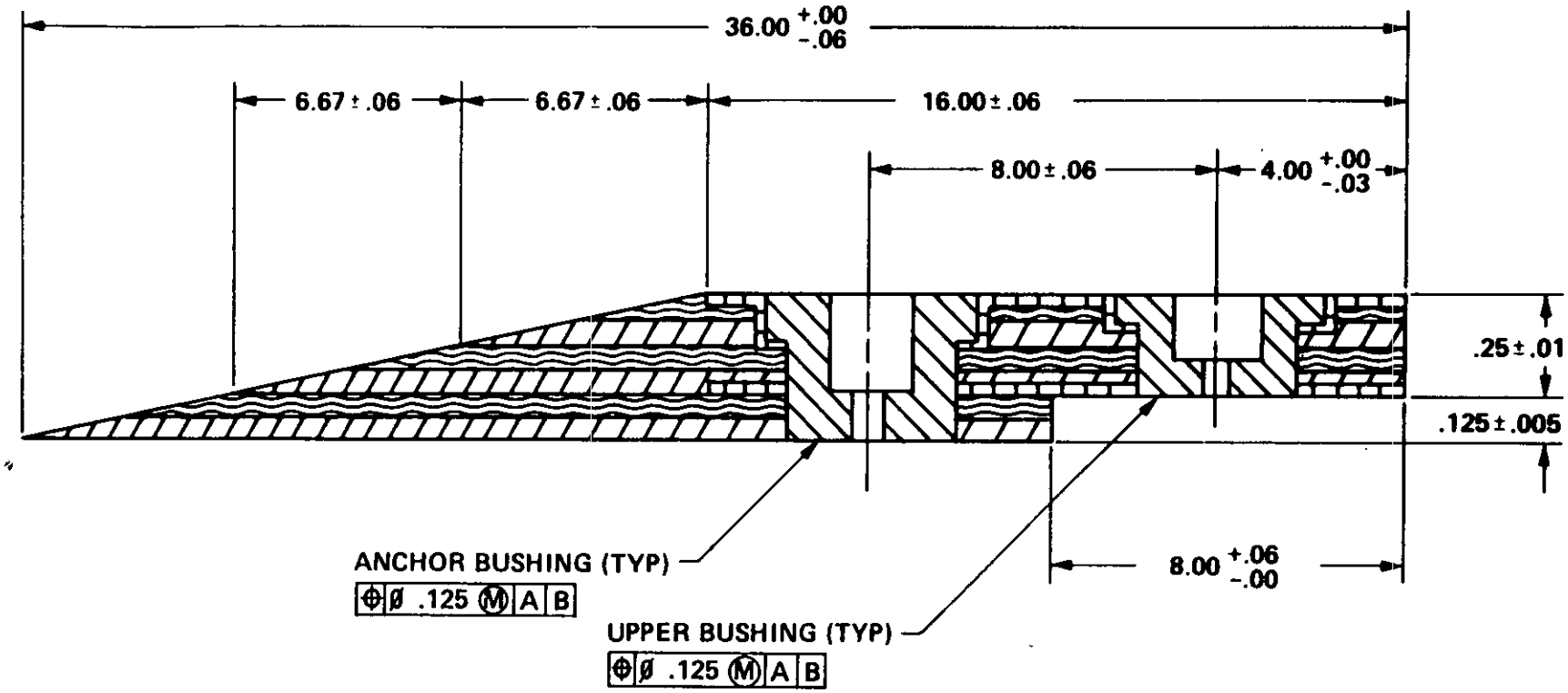
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NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. UNTOLERANCED DIMENSIONS ARE BASIC.
3. FIGURE IS NOT TO SCALE.

FIGURE 7. FRP airfield matting panel (RTM method), ramp panel.

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WOVEN ROVING

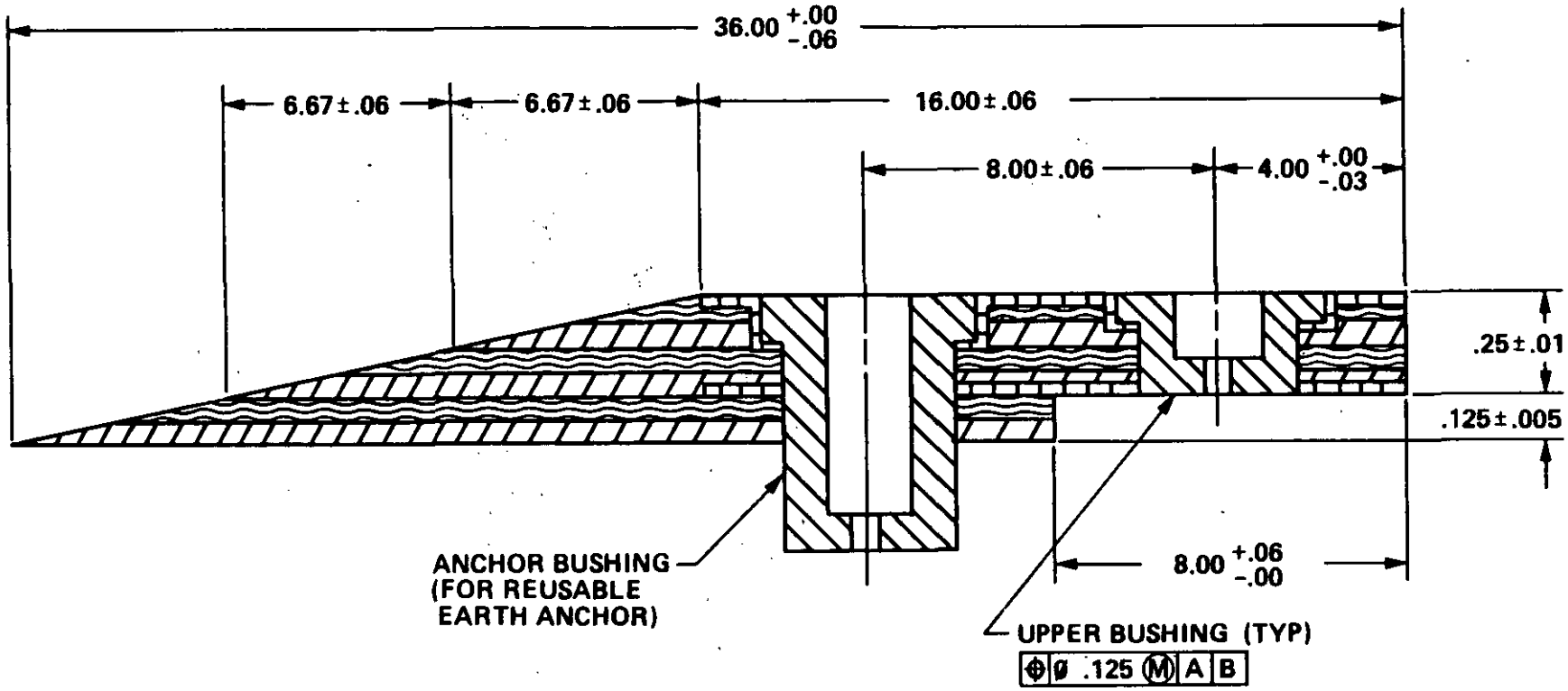
CONTINUOUS STRAND

WOVEN ARAMID

NOTE:
1. ALL DIMENSIONS ARE IN INCHES
2. FIGURE IS NOT TO SCALE.

FIGURE 8. Ramp panel, regular anchor bushing, Section C-C.

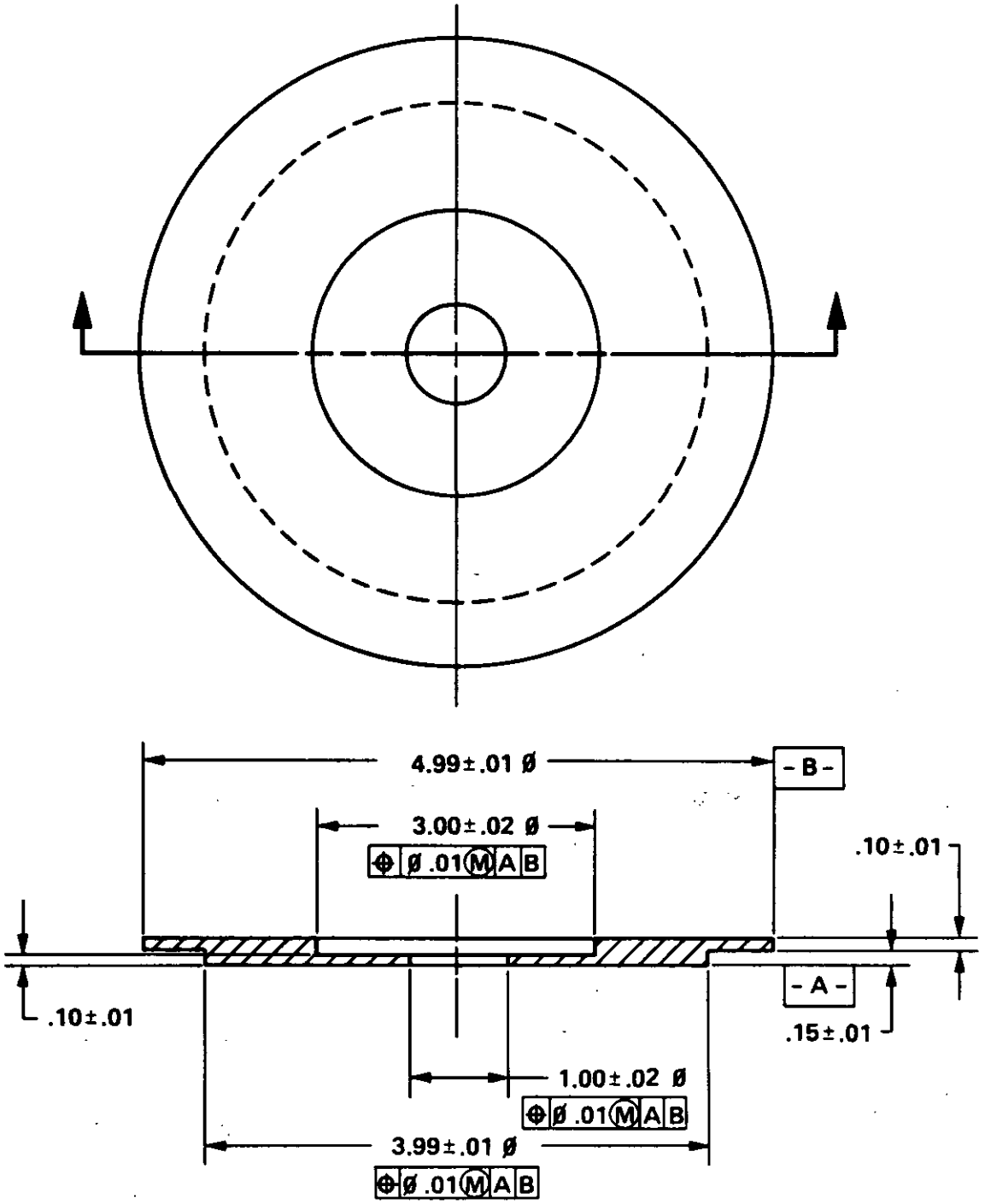
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NOTE:
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2. FIGURE IS NOT TO SCALE.

FIGURE 9. Ramp panel, earth anchor bushing, Section C-C.

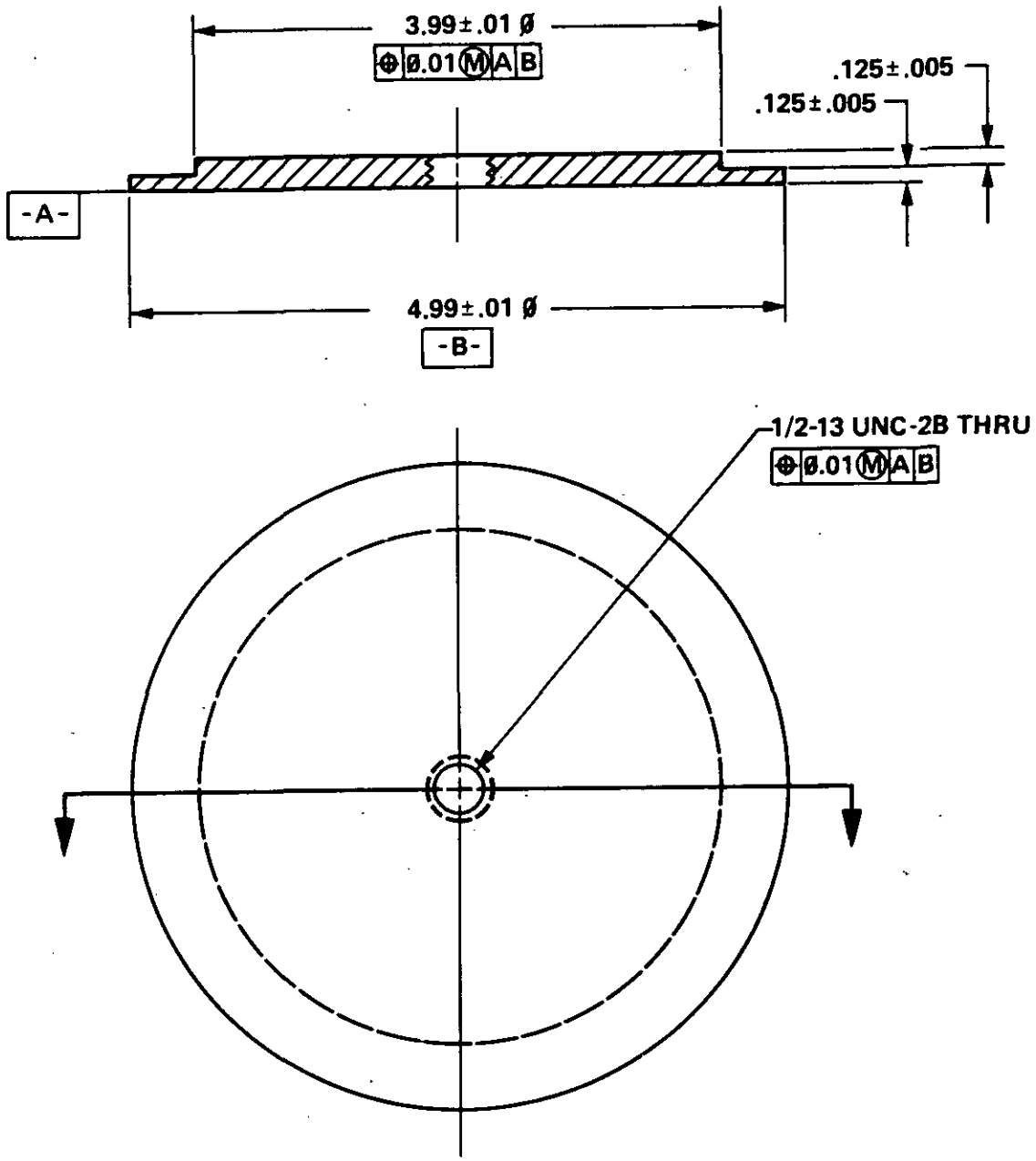
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NOTE:
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2. FIGURE IS NOT TO SCALE.

FIGURE 10. Upper bushing detail.

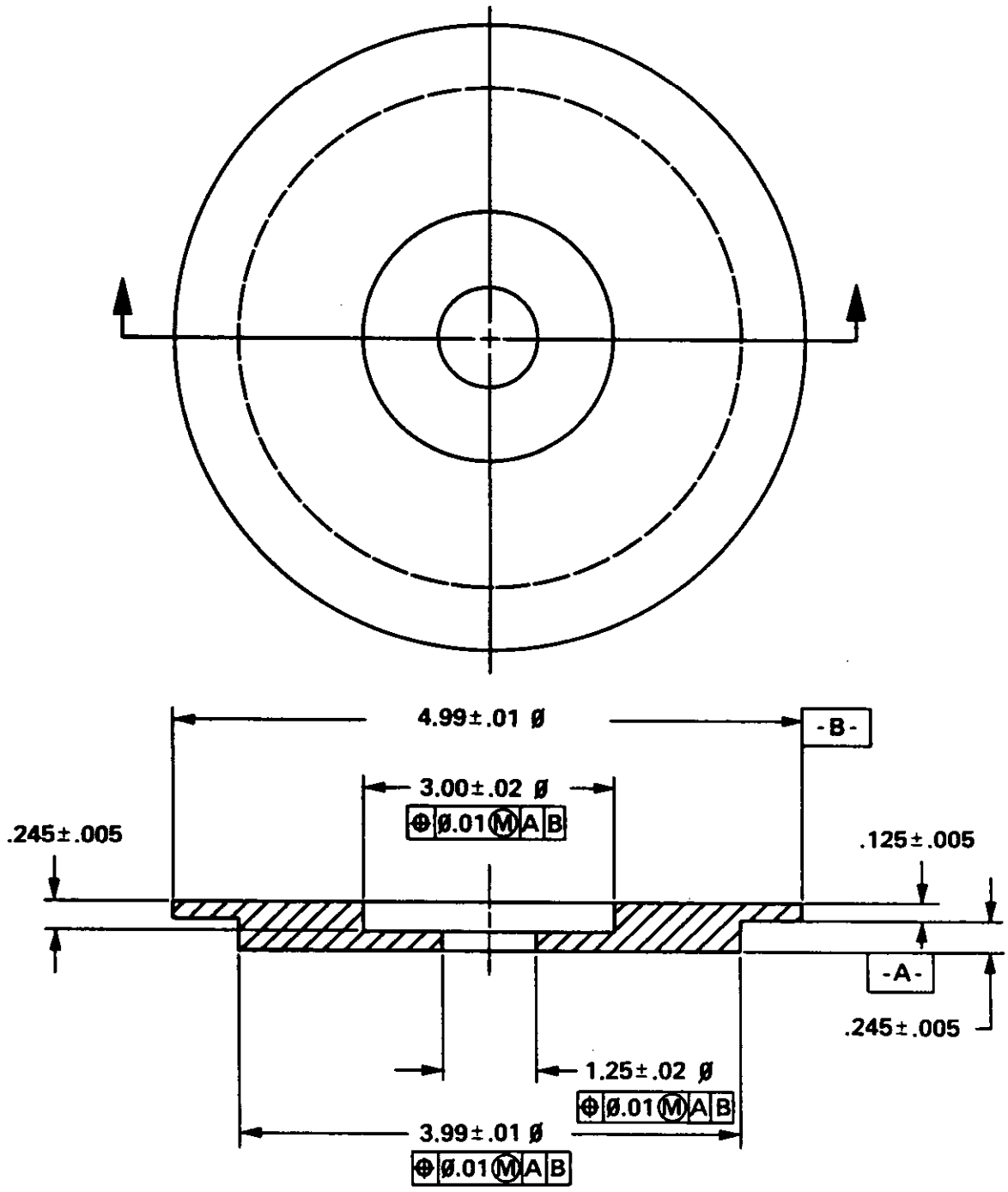
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NOTE:
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2. FIGURE IS NOT TO SCALE.

FIGURE 11. Lower bushing detail.

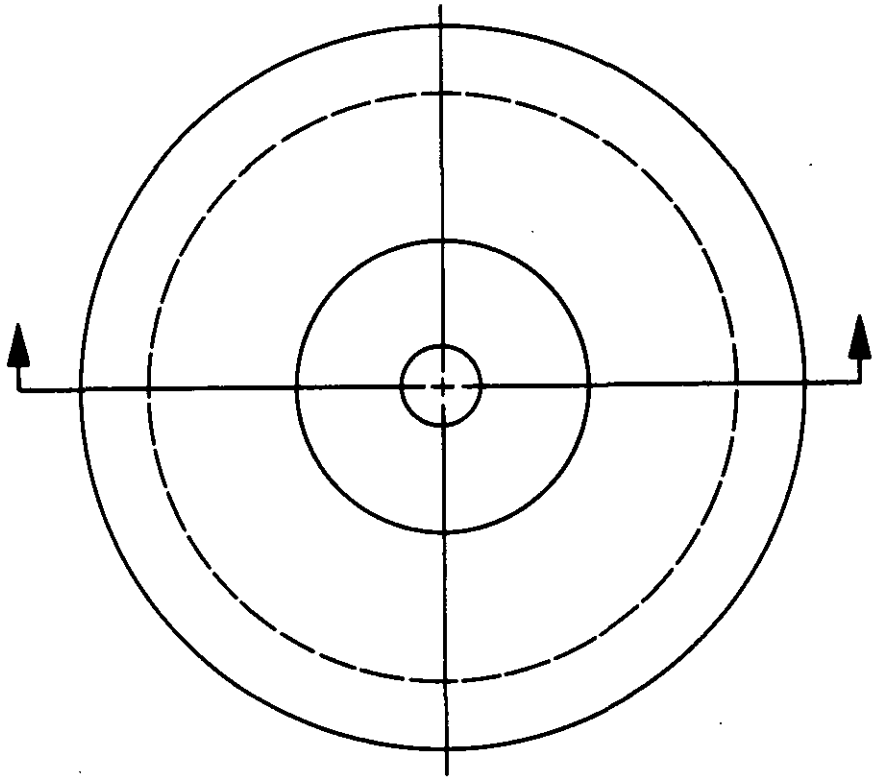
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NOTE:
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2. FIGURE IS NOT TO SCALE.

FIGURE 12. Anchor bushing detail.

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NOTE:
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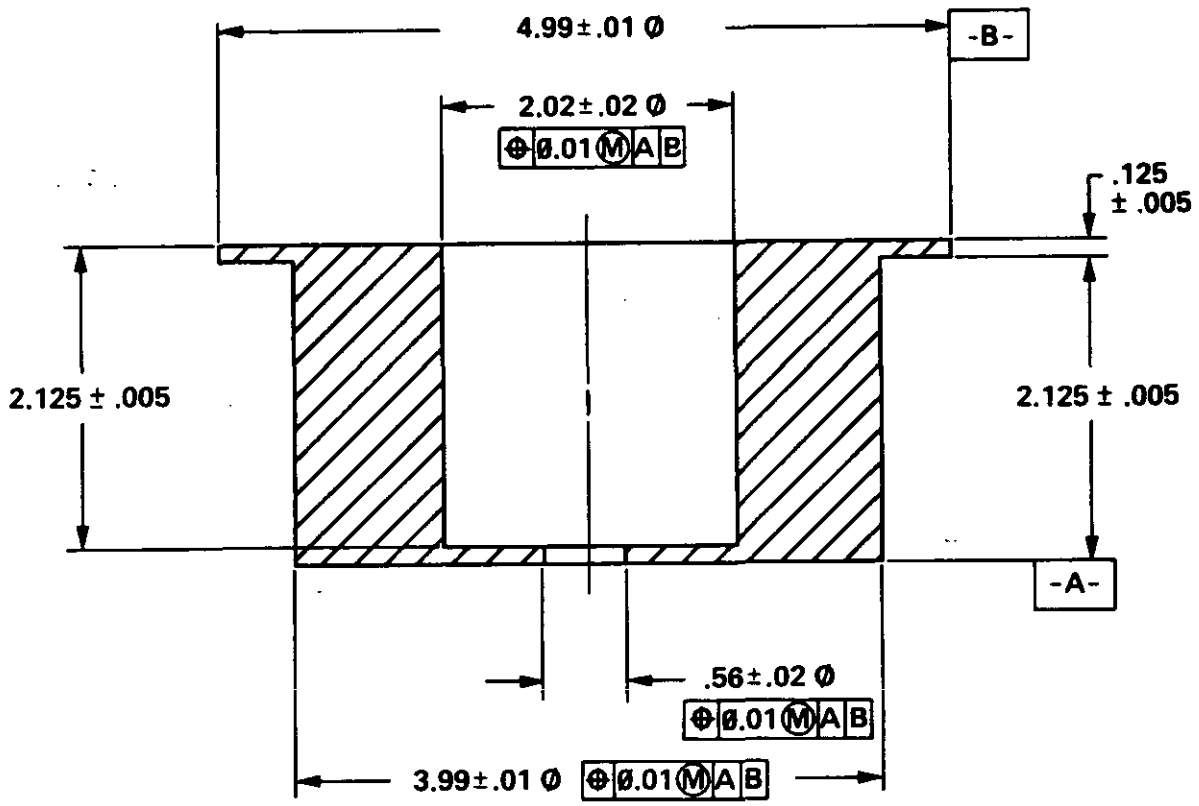


FIGURE 13. Anchor bushing detail (earth anchor).

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NOTES:
1. BREAK ALL SHARP EDGES (TYP. SHEETS 1-3)
2. MATERIAL: AISI C1144 STEEL (TYP. SHEETS 1-3)
3. ALL DIMENSIONS ARE IN INCHES.
4. FIGURE IS NOT TO SCALE.

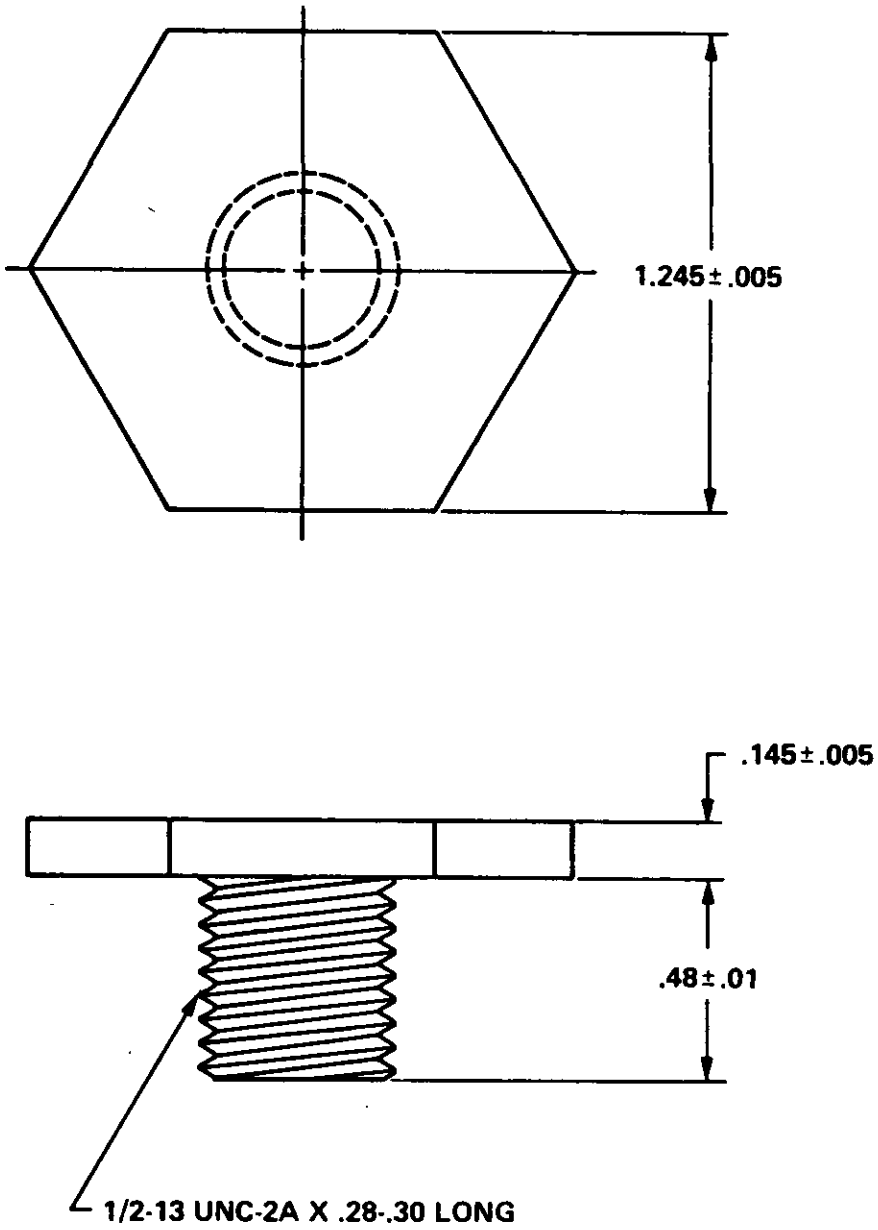
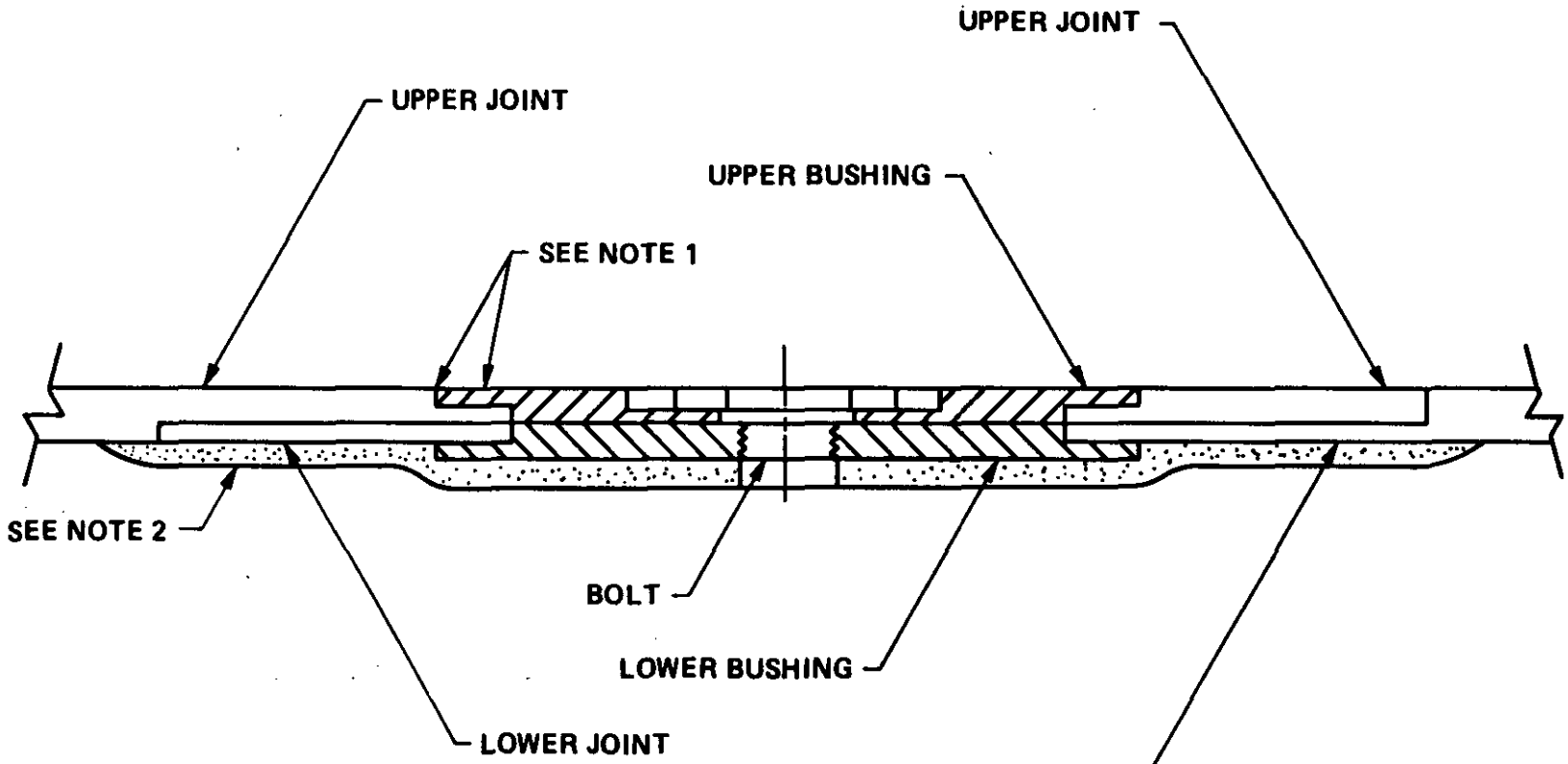


FIGURE 14. Bolt.

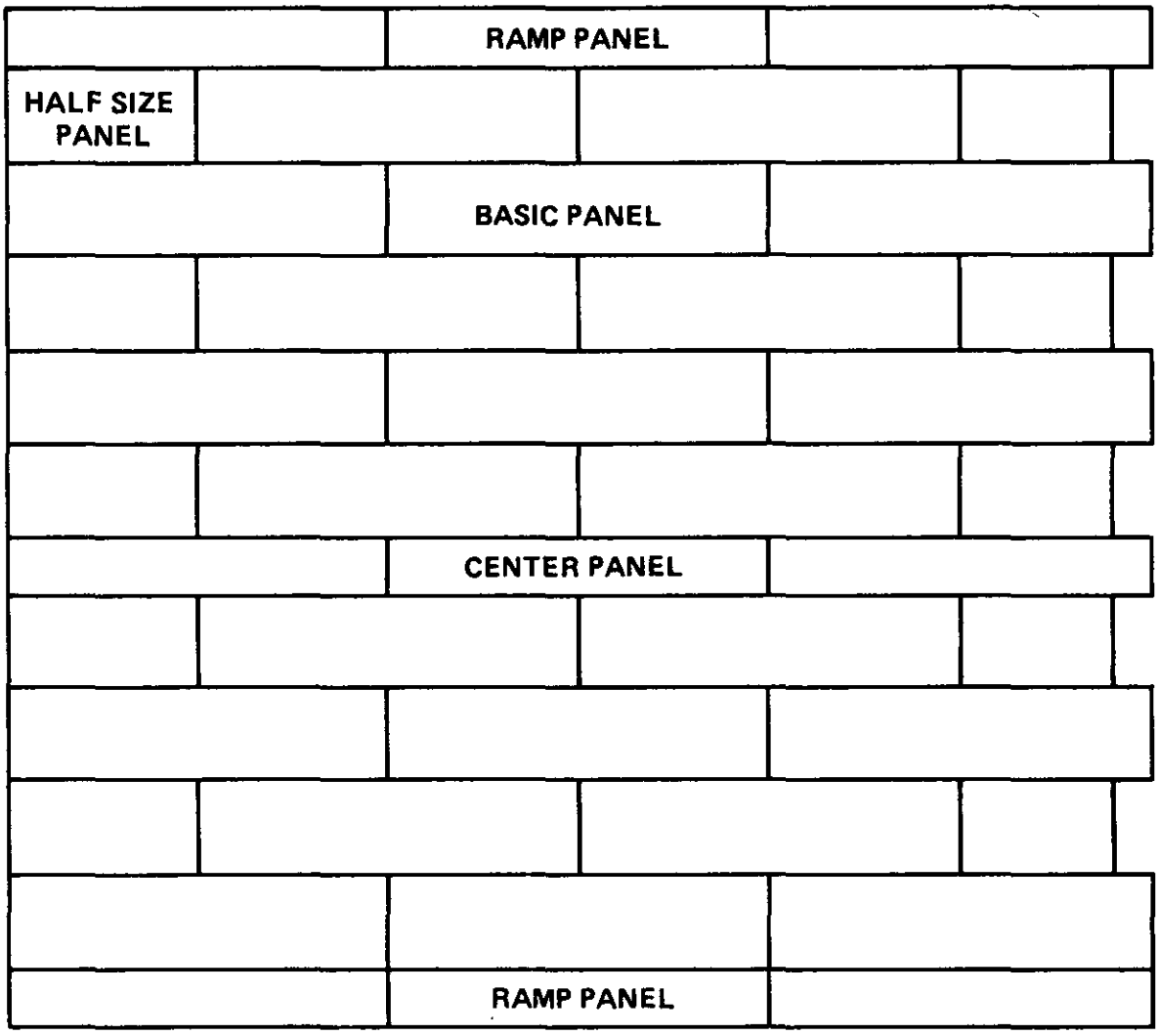
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NOTE:
1. IT IS IMPERATIVE THAT THE UPPER LAYER OF WOVEN ARAMID ON THE UPPER JOINT CONTINUE UNDERNEATH THE FLANGE OF THE UPPER BUSHING TO THE 4" DIAMETER OF THE BUSHING.
2. FIBERGLASS COVER UNDER BOTTOM BUSHING EXTEND MINIMUM OF 3" BEYOND BUSHING CIRCUMFERENCE.
3. ALL DIMENSIONS ARE IN INCHES.
4. FIGURE IS NOT TO SCALE.

FIGURE 15. Bushing/Joint assembly, Cross Section.

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NOTE:
1. FIGURE IS NOT TO SCALE.

FIGURE 16. Assembled FRP cover, Plan View.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P29242(YD)	PANELS, FIBER - REINFORCED PLASTIC FOR RAPID RUNWAY REPAIR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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